

1

Data Preparation

TranStats Flight Records

≈ 61K Domestic US Flights
(including 127 diverted flights)

Feature Engineering

Randomly missing features

Time features to UTC

Remove Relational Features

Handle Diverted Flights

Preprocessed Flight Records

2

Synthetic Data Generation

Generative Models

Gaussian Copula

TVAE

CTGAN

CopulaGAN

Multi-objective Tuning

- Realism
- Statistical Similarity
- Fidelity
- Utility

Synthetic Flight Diversions

Augmentation

3

Quality Evaluation

Performance Metrics

Synthetic vs. Real

Realism

Diversity

Operational Validity

Statistical Similarity

Fidelity

TATR vs. TRTR

Utility

Only Diversions

Only Diversions

Only Diversions

Full Records
Full Records